(A) are used for infective endocarditis. Surgical intervention for tricuspid regurgitation (C, D) is reserved for patients with right heart failure refractory to medical treatment.

4. When comparing valvular disease from carcinoid syndrome (CS) versus rheumatic heart disease (RHD), which of the following is true?

- **A.** ○ Aortic valve disease is common with CS.
- **B.** ○ CS more commonly causes stenosis than regurgitation.
- **C.** ● RHD more commonly causes mitral valve disease than tricuspid disease.
- **D.** ○ RHD causes stenosis early but regurgitation in chronic cases.

Hide Explanation

The correct answer is (C). RHD more commonly affects left heart valves because of the higher turbulent flow, but it can also affect right heart valves. CS mostly affects right heart valves, because serotonin is mostly eliminated from the circulation in the lungs (A). Early valve disease from both CS and RHD is more likely to impair precise valve closure early, with stenosis being a later result of advanced scarring. (B, D)

How would you rate this Brick?

ALICE L WALTON SCHOOL OF MEDICINE

Tricuspid Regurgitation and Tricuspid Stenosis

Last updated December 26, 2025 · 17 min read · ScholarRx

Adapted by: Diptiman Bose, Patrick Brooks

Editors: Jordan Finn, Russell Johnson

Adapted from: Tricuspid Regurgitation and Tricuspid Stenosis

Listen to this Brick
0:00 / 27:01 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the etiologies and pathophysiology of tricuspid regurgitation.
2. Describe the clinical presentation, diagnosis, and management of tricuspid regurgitation.
3. Describe the … …uspid stenosis.

ALICE L WALTON SCHOOL OF MEDICINE

Mitral Stenosis

Last updated February 20, 2026 · 17 min read · CC BY-SA 4.0

Adapted by: Diptiman Bose, Patrick Brooks, Phil Riley, Ian Murray

Editors: Chelsey Deel, Jordan Finn, Russell Johnson

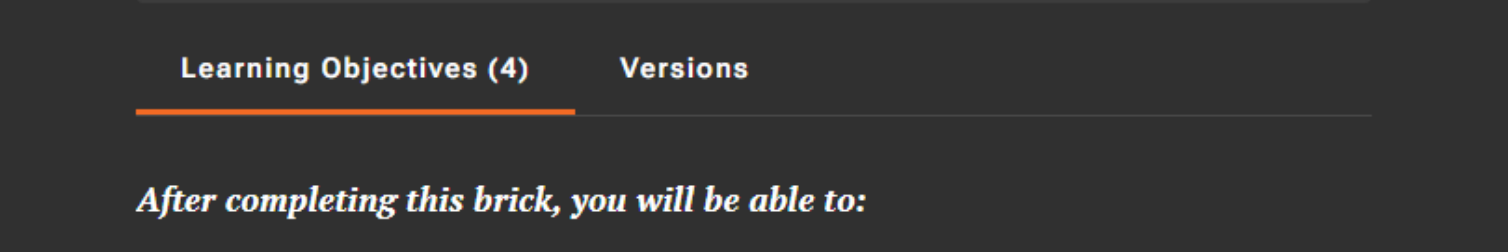
Adapted from: Mitral Stenosis

Listen to this Brick
0:00 / 26:02 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Define mitral stenosis and its clinical presentation.
2. Explain the etiology, pathophysiology, and complications of mitral stenosis.
3. Explain the diagnosis of mitral stenosis.
4. Describe the



QUIZ

Figure 2 Mitral valve anatomy

The anterior and posterior valve leaflets control the passage of blood from the LA to the LV - the *two leaflet* structure allows tighter closure during the strong LV contractions (the *three leaflet* aortic valve allows more rapid and complete opening for the LV ejection). The annulus is a ring-like structure that holds the leaflets in place. The **papillary muscles** are attached to the heart wall muscle and contract to pull back the **chordae tendineae** into a taut position. These **hold the mitral valve (and tricuspid valve) closed during the high pressure of ventricular systole and prevent the leaflets from prolapsing backward into the atria.** (Aortic and pulmonary valves *open* during the strong ventricular contractions, so they are freestanding without papillary/chordae apparatuses.)

The papillary muscles do not open or close any heart valves. All heart valves open and close passively due to changes in pressure.

How Do Patients With Mitral Stenosis Present? (LO 1)

Patients with MS most commonly present with **exertional dyspnea** and decreased exercise to… …es slowly, patients

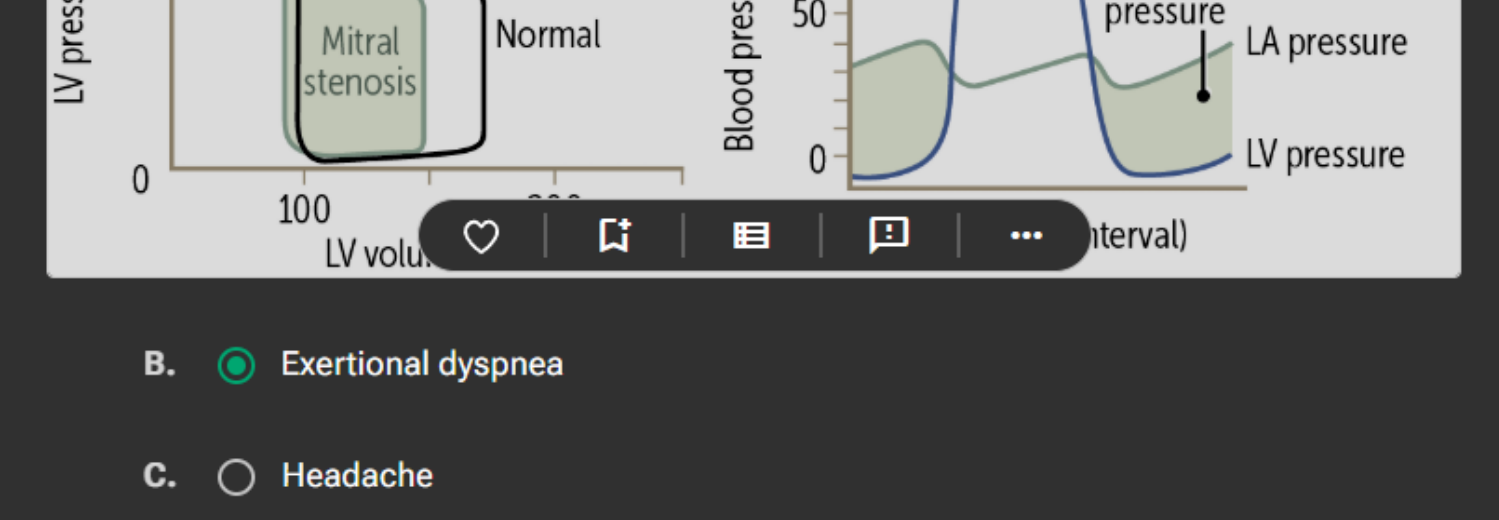
valve is the most common heart valve involved in RHD. Rheumatic fever is an inflammatory condition that can develop after inadequate treatment of streptococcal infections caused by group A streptococcus (GAS). It triggers an immune response to the antigens on GAS which are very similar to the structure of valvular tissue, resulting in autoimmune attack of the heart valves. Although the prevalence of rheumatic heart disease in developed nations like the United States is on the decline, an estimated 1 in 100,000 patients still contract this disease. Initially, rheumatic fever may present with Mitral Regurgitation. However, over a period of years to decades, chronic inflammation of the mitral valve leads to thickened, calcified leaflets of the valve with fibrin deposits on the cusps causing stenosis. Severe inflammation results in **commissural fusion** - the leaflets fuse together at the commissures. As a result, the stenotic mitral valve from rheumatic disease develops a characteristic slit-like, curved opening that resembles a fish mouth (Figure 4).

- **Calcification of Annulus.** In older patients calcium can deposit along the annulus. Annular calcification more commonly interferes with leaflet closure and results in Mitral Regurgitation. In advanced cases it can cause Mitral Stenosis by limiting the annulus's ability to expand during diastole and by restricting the movement of the valve leaflets (Figure 5). The mitral valve *annulus* is much less often calcified than the aortic valve, where calcification affects valve *leaflets*, because the aortic valve has a smaller diameter, higher rate of peak flow, and more turbulence.

What is the most common cause of Mitral Stenosis?

Pathophysiology of Mitral Stenosis (LO 2)

The main problem in MS is that the stenotic valve causes an increase in both LA volume and pressure during diastole (Figure 6, right).



- **B.** ● Exertional dyspnea
- **C.** ○ Headache
- **D.** ○ Hoarseness
- **E.** ○ Palpitations

Hide Explanation

The correct answer is exertional dyspnea (B). Mitral Stenosis (MS) leads to backup of blood in the left atrium, causing excess pressure on the lungs. This causes pulmonary hypertension, and patients become short of breath with exertion or exercise. Cough (A) can be a symptom of MS when patients develop pulmonary hypertension or pulmonary edema, but it is less common than dyspnea. Headache (C) is not typically associated with MS. Hoarseness (D) is a rare complication of MS that occurs when the left atrium dilates to the extent that it compresses the laryngeal nerve, which controls the voice. Palpitations (E) can occur with MS if patients develop atrial fibrillation, but they are less common than dyspnea.

3. Which of the following is the preferred treatment option to treat the underlying condition causing mitral stenosis?

- **A.** ○ β-Blockers
- **B.** ○ Calcium

References

- Dima C. Mitral stenosis. Medscape. Updated December 8, 2021. Accessed September 27, 2022. https://emedicine.medscape.com/article/155724-overview#a6
- *StatPearls*. StatPearls Publishing. Updated July 8, 2022. Accessed September 27, 2022. https://www.ncbi.nlm.nih.gov/books/NBK430742/

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Define mitral stenosis and its clinical presentation.~~

02 ☑ ~~Explain the etiology, pathophysiology, and complications of mitral stenosis.~~

03 ☑ ~~Explain the diagnosis of mitral stenosis.~~

04 ☑ ~~Describe the management of mitral stenosis.~~

Objective Confidence: 4/4

What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Mitral Stenosis

Last updated February 20, 2026 · 17 min read · CC BY-SA 4.0

Adapted by: Diptiman Bose, Patrick Brooks, Phil Riley, Ian Murray

Editors: Chelsey Deel, Jordan Finn, Russell Johnson

Adapted from: Mitral Stenosis

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0:00 / 26:02 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Define mitral stenosis and its clinical presentation.
2. Explain the etiology, pathophysiology, and complications of mitral stenosis.
3. Explain the diagnosis of mitral stenosis.
4. Describe the

ALICE L WALTON SCHOOL OF MEDICINE

Infective Endocarditis

Last updated March 9, 2026 · 20 min read · CC BY-SA 4.0

Adapted by: Rhome Hughes, Chelsey Deel, Stephen Nix, Diptiman Bose, Ray Brown, Lori Garman

Editors: Jordan Finn, Russell Johnson

Adapted from: Infective Endocarditis

Listen to this Brick
0:00 / 32:30 1x

Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Recognize the three types of infective endocarditis (IE).
2. Differentiate the clinical presentation of IE.
3. Explain the pathophysiology of IE.
4. Interpret clinical findings to properly diagnose IE.
5. Outline the

What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Infective Endocarditis

Last updated March 9, 2026 · 20 min read · CC BY-SA 4.0

Adapted by: Rhome Hughes, Chelsey Deel, Stephen Nix, Diptiman Bose, Ray Brown, Lori Garman

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Adapted from: Infective Endocarditis

Listen to this Brick
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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Recognize the three types of infective endocarditis (IE).
2. Differentiate the clinical presentation of IE.
3. Explain the pathophysiology of IE.
4. Interpret clinical findings to properly diagnose IE.
5. Outline the

Injury to the endocardium causes fibrin and platelets to form a thrombus at the injury site. Infection occurs when bacteria are introduced into the thrombus through the blood. For example, bacteria initially enter the blood through dental procedures (including tooth brushing), with injectable drug use, or simply by chewing food. Bacteria will then adhere to and colonize the thrombus, forming vegetations composed of platelets, fibrin, and bacteria.

The three types of endocarditis are divided into two categories:

- **Native valve endocarditis,** which involves the patient's natural valves. If symptom onset is rapid, the condition is classified as **acute IE.** If symptom onset is gradual, usually with chronic nonspecific symptoms, the condition is considered **subacute IE.**
- A separate category is **prosthetic valve endocarditis,** when installed valves are infected after the native valves have been surgically replaced.

How Do Patients with Infective Endocarditis Present? (LO #2)

The most common symptom of IE is **fever** (present in 90% of patients), but this is not a very specific finding. Patients may also experience altered mental status, vision changes, shortness of breath, lower extremity swelling, joint and back pains, and fatigue and lethargy, but these will all vary; some patients present only with fever. Unlike many other causes of fever, patients with IE often present without any localizing symptoms (e.g., cough, dyspnea, skin rashes, or urinary frequency or dysuria) to suggest the origin of the infection. So, IE is often detected on an evaluation of "fever of unknown origin."

As we'll see in the di… …s can help us diagnose IE, especially a new heart murmur (present in 85% of patients)

Why are patients with prosthetic valves predisposed to endocarditis?

Pathogens in Special Populations

Some patient populations classically develop endocarditis in association with specific pathogens.

Patients with Congenital or Acquired Valvular Disease. Those with underlying valve disease such as rheumatic heart disease are at risk for subacute IE caused by the **viridans group streptococci or *Enterococcus*.** Bacteremia in associated with these bacteria typically only produces endocarditis on damaged cardiac valves. These bacteria are considerably less virulent than *S. aureus*. Their lower virulence explains why most of these cases have subacute presentations.

Patients Who Injec… susceptible to *S. aureus* endocarditis due to direct inoculation of the bacteria from the skin